

Sounak Chakraborty

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SUMMARY

Detail-oriented MCA graduate (2027) with strong foundations in software development, system design, and web technologies. Proficient in C, Java, Python, and JavaScript, with hands-on experience building production-ready full-stack applications — from AI-powered fitness platforms to intelligent API gateways with real-time analytics, ML-based anomaly detection, and distributed systems patterns like circuit breaking and adaptive rate limiting.

PROJECTS

Gatekeeper – Intelligent Adaptive API Gateway

2026

- Engineered a high-performance, self-healing API gateway using **Node.js**, **Express**, **MongoDB**, **Redis**, **React 19**, and **Socket.io** with Docker-based deployment.
- Implemented adaptive **Redis-backed rate limiting** (token bucket), a three-state **circuit breaker** (CLOSED/OPEN/HALF_OPEN), and automated **DDoS protection** with per-route request thresholds.
- Built weighted round-robin **load balancing** with periodic health probes (0–100 health score), automatic exclusion of unhealthy backends, and W3C-compatible **distributed tracing**.
- Integrated an **ML-based anomaly detection** module using a **Linear Regression model** to identify abnormal traffic patterns and proactively flag suspicious client behavior.
- Developed a real-time **React dashboard** with Socket.io WebSocket streaming for live traffic, latency, circuit state, and alerts, with SSE fallback for log streaming.
- Secured the system with **better-auth** (JWT sessions), SHA-256-hashed **API key authentication**, **Helmet.js** headers, **CORS**, and role-based admin access control.

Fitoholic – Full-Stack AI-Powered Fitness Tracker

2025

- Built a comprehensive, production-ready fitness tracking web app tailored for Indian users using **Spring Boot (Java 21)** and **React 18**.
- Implemented secure **JWT-based authentication** with password reset, premium access gating, and role-based admin control.
- Integrated **Razorpay API** for seamless one-time payments and enabled premium users to access “Fitto,” an AI fitness advisor powered by **Google Gemini API**.
- Enabled real-time tracking of Steps, Calories, Water Intake, and Weight with animated chart visualizations using **Framer Motion**.
- Developed a secure Admin Dashboard with user monitoring, stats overview, and read-only access to fitness logs.
- Designed a responsive UI with **Tailwind CSS**, Light/Dark theming, and dashboard routing via **React Router DOM**.
- Used **Spring Security 6**, **MySQL 8**, **Hibernate (Spring Data JPA)**, and SMTP mail integration for secure notifications and email services.

TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript, HTML/CSS

Frameworks/Libs: Spring Boot, React 18/19, Tailwind CSS, Framer Motion, Express.js, Socket.io

Database & ORM: MySQL 8, PostgreSQL, MongoDB, Redis, Spring Data JPA (Hibernate), Mongoose

Dev Tools: Git, Maven, Docker, Eclipse, Visual Studio Code, Postman

API Integrations: Razorpay API, Google Gemini API, Fetch API, SMTP Mail

Other: JWT, Spring Security, RESTful APIs, Circuit Breaker, Rate Limiting, Responsive Design, Accessibility (ARIA)

Operating Systems: UNIX, Windows, Linux

Soft Skills: Problem solving, Teamwork, Communication, Critical thinking

EDUCATION

Christ University

MCA

Bangalore

2025 – 2027

Techno Main Salt Lake

BCA 7.8 CGPA

Kolkata

2022 – 2025

South Point High School

Commerce 86%

Kolkata

2006 – 2022